



# MANIPAL ACADEMY OF HIGHER EDUCATION

**Manipal Institute of Technology, Bengaluru**  
BTech VI Semester  
End Semester Examination - April / May -2025  
**BIG DATA ANALYTICS [IT-4410]**

**Marks: 50**

**Duration: 180 mins.**

## PART-A

**Answer all the questions.**

- 1) Discuss the life cycle of Big data analytics. (5)
  - A)
  - B) An e-commerce company collects massive volumes of customer interaction data, including browsing history, purchases, and reviews. The business team wants to improve customer satisfaction and sales by identifying popular products, understanding product performance issues, forecasting demand, and optimizing pricing and inventory. Apply your understanding of the types of Big Data Analytics to this scenario. Demonstrate how each type-when correctly applied-can help the company address its goals with suitable justifications (5)
- 2) A data analytics firm processes large datasets daily and stores them in an HDFS cluster for distributed analysis. A data engineer uploads a 1GB log file to HDFS. During the write operation, the file is divided into blocks and replicated across multiple nodes.
  - A) However, midway through the upload, one of the DataNodes storing a block of the file crashes. Determine how HDFS ensures that the file is successfully written despite the failure. In addition articulate the key steps involved in the write operation with diagram, including how fault tolerance is maintained. (5)
  - B) Consider a collection of literature survey made by a researcher in the form of a text document with respect to cloud and big data analytics. Analyze Using Hadoop and Map Reduce, write a program to count the occurrence of pre dominant key words. Consider pre dominant words as Big Data, Hadoop, MapReduce (5)
- 3) A logistics company is facing delays and inefficiencies with its current batch processing system built on traditional MapReduce. To enable faster analytics and better system performance, the company plans to migrate to Apache Spark. Apply your understanding of Apache Spark to propose a solution that improves system performance and scalability. Consider how Spark's in-memory computation, multi-language support, fault tolerance, and unified engine would benefit this logistics use case. (5)
  - A)
  - B) A retail company wants to monitor user activity in real time by analyzing streaming clickstream data from its e-commerce platform. Each data record includes user ID, timestamp, and visited page URL. The company needs to compute the number of clicks per page every minute and also keep a cumulative count of total clicks per page. Apply your understanding of DStreams and Spark Streaming transformations to design a solution that meets these requirements. (5)
- 4) Determine the storage strategy followed with relevant examples of databases where Riak and CouchDB applications are built upon (5)

A)

B) Correlate below SQL commands to its equivalent MongoDB commands

a). UPDATE stud SET marks\_dmsa = 50 WHERE class = 'TE';

b). SELECT \* FROM stud WHERE \_id = '5063114bd386d8fadbd6b004' AND Name = 'Reena' AND Rno = 15;

c) SELECT \* FROM students WHERE course NOT IN ('Math', 'Science');

d) Select \* from users order by name desc;

e) select count(\*) from stud where rollno=2;

(5)

5) Describe Unsupervised and Reinforcement learning algorithms in detail.

(5)

A)

B) Implement K Means algorithm using Pyspark assuming a suitable dataset.

(5)

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